CPT205
Computer Graphics

Name: Yifei Li

ID: 2364053

Degree programme: DMT

1、Card functions and related graphic technologies

(1) The photo frame and the filling mode switch between each other. The iconic buildings of XJTLU, which are constructed with 41 pictures in the card: the central building library, the FB Foundation building, the White House, as well as the trees on the ground and around, all adopt this mode, enhancing the interactivity and visual impact of the card surface. Regarding the code:

(a) Define the wireframes to facilitate the subsequent switching between wireframes and fillings.

```
bool isWireframe = true;// Wireframe pattern definition
```

(b) Switch between wireframes and fillings through the left mouse button interaction.

```
void myMouse(int button, int state, int x, int y) {// Mouse interaction
    if (button == GLUT_LEFT_BUTTON && state == GLUT_DOWN) {// triggered by left-click //of the mouse
    isWireframe = ! isWireframe;   // Switch wireframe/Fill mode
```

(c) The switching representation of graphic wireframes and fillings.

```
void drawShapeLightGray(GLfloat vertices[][2], GLint indices[], int count) {
    if (isWireframe) drawOutline(vertices, indices, count);   // Wireframe mode: Only draw //wireframes
    else {// Fill in mode: Fill in first and then draw the wireframe
    drawFillBackgroundColor(vertices, indices, count);
    drawOutline(vertices, indices, count);
  }
}
```

(2) The windmill rotates and switches. The stop, transformation, acceleration and deceleration of the windmill are controlled by special keys. Regarding the code:

(a) The timer update function for the windmill's rotation.

```
void OnTimer(int value) {
    r += step;                    // Update the rotation Angle based on the step size
    if (r >= 360) r = 0;      Angle normalization (reset to 0 when exceeding 360 degrees)
    else if (r <= 0) r = 359;
    glutTimerFunc(time_interval, OnTimer, 1);     // Loop trigger timer
}
```

(b) Keyboard special key control logic.

```
void keyboard_input(unsigned char key, int x, int y) {
switch (key) {
case 'q': case 'q': // Exit the program
exit(0);
break;
case 'f': case 'f': // Switch the rotation direction
step = -step;
break;
case 's': case 's': // Stop rotating
step = 0;
```

```
break;
case 'r': case 'R': // Slow down
if (step >= -15) // limits the minimum step size
step--;
break;
case 't': case 'T': // Acceleration
if (step <= 15) // limits the maximum step size
step++;
break;
}
}
```

(3) Drawing of graphics. Regarding the code:

(a) The fan blades are drawn in a triangular pattern.

```
glBegin(GL_TRIANGLES);
glVertex2f(0.0f, 0.0f);
glVertex2f(0.0f, 100.0f);
glVertex2f(90.0f, 130.0f);
glEnd();
```

(b) The drawing of the building utilized polygon drawing.

```
glBegin(GL_POLYGON);
    for (int i = 0; i < count; i++) {
        glVertex2f(vertices[indices[i]][0], vertices[indices[i]][1]);
    }
    glEnd();
}
```

(c) The drawing of the tree crown uses the drawing of line loops.

```
glBegin(GL_LINE_LOOP);
for (int i = 0; i < 36; ++i) {
    float angle = static_cast<float>(i * 10.0 * M_PI / 180.0);
    glVertex2f(15.0f * cos(angle), 15.0f * sin(angle));
}
glEnd();
```

(d) The drawing of the tree trunk uses the drawing of lines.

```
glBegin(GL_LINES);
glVertex2f(502.0f, 75.0f);
glVertex2f(502.0f, 35.0f);
glEnd();
```

(e) The drawing of clouds uses the drawing of triangular sectors.

```
glBegin(GL_TRIANGLE_FAN);
glVertex2f(0.0f, 0.0f);
for (int j = 0; j <= 36; j++) {
    float angle = static_cast<float>(j * 10.0 * M_PI / 180.0);
    float x = cos(angle) * 50.0f * sizeFactor;
    float y = sin(angle) * 30.0f * sizeFactor;
```

```
    glVertex2f(x, y);
}
glEnd();
```

(4) Text Drawing. Regarding the code:

(a) Render the font by calling the OpenGL freeglut APIs.

```
HDC hDC = wglGetCurrentDC();
HFONT hOldFont = (HFONT)SelectObject(hDC, hFont);
fontListBase = glGenLists(MAX_CHAR);
wglUseFontBitmaps(hDC, 0, MAX_CHAR, fontListBase);    // Convert the system font to the //OpenGL
renderable format
```

(5) Move and zoom. By moving and zooming, the constructed graphics can reach the target size and position. Regarding the code:

(a)Static movement and zooming.

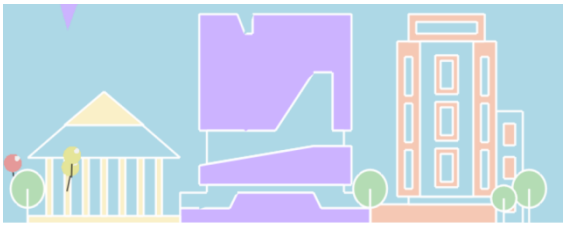
```
glTranslatef(TRANS_X, TRANS_Y, 0.0f);
glScalef(SCALE_FACTOR, SCALE_FACTOR, 1.0f);
```

(b)Dynamic movement: Add dynamic movement states to clouds and balloons to enable them to move repeatedly along fixed routes.

```
void updateClouds(int value) {
int windowWidth = glutGet(GLUT_WINDOW_WIDTH);
for (auto& cloud : clouds) {
cloud.x += cloud.speed;    // Move along the X-axis (core movement logic)
if (cloud.x > windowWidth + cloud.size) {
cloud.x = -50.0f - rand() % 350; // Reset the x-coordinate
cloud.y = 350.0f + rand() % 120; // Random Y-coordinate
}
}
glutPostRedisplay();
glutTimerFunc(16, updateClouds, 0);    Refresh once every 16ms
}
```

2、Operation Instructions for Greeting cards

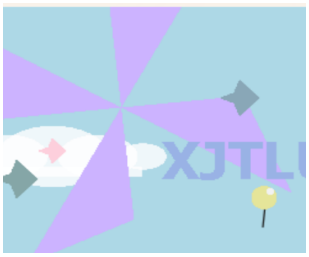
(1) With the left mouse button, the iconic buildings of XJTLU, namely the Central Library, the FB Foundation Building and the White House, can freely switch between wireframe and fill modes.



(2) The left mouse button can be used to freely switch between the wireframe and fill modes for the ground and trees.

(3) The left mouse button can be used to switch the colors of the windmill (white and purple) and the text (white and blue).

(4) Press the 'Q' key to close the page;
The 'F' key controls the windmill to rotate in the opposite direction;
The 'S' key controls the windmill to stop;
The 'R' key controls the windmill to accelerate;
The 'T' key controls the windmill to slow down.



(5) The clouds in the card move repeatedly to the right along the X-axis, the balloon moves repeatedly from the lower left to the upper right along the Y= X-axis, and the stars twinkle repeatedly.

(6) Overall effect drawing.